BINATIONAL MASTER'S DEGREE IN INDUSTRIAL INFORMATICS

DIGITALIZATION PROPOSAL FOR COLD ROOMS IN A FRUIT AND VEGETABLE MARKET

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Introduction

Cold rooms play a crucial role in preserving perishable goods at the Fruit and Vegetable Market of Santa Fe. These storage facilities operate 24/7 to maintain the required temperature and humidity levels for fresh produce. However, the market faces several operational challenges, particularly in energy efficiency, monitoring, and maintenance.

Many of the cold rooms use outdated technology and lack real-time monitoring capabilities, making it difficult for operators to optimize energy consumption and predict failures. This lack of digitization results in high energy costs, operational inefficiencies, and frequent equipment breakdowns.

To address these issues, we propose a digitalization strategy that involves implementing an Asset Administration Shell (AAS) for one cold room as an example. This digital model will serve as a standardized representation of the cold room, allowing for real-time monitoring, predictive maintenance, and data-driven decision-making.

Through this proposal, we aim to:

- Identify the key challenges that cold rooms face.
- Define the requirements for digitalization.
- Show how AAS can be implemented to improve efficiency.
- Provide a roadmap for future implementation at scale.

By adopting AAS technology, cold rooms can transition from manual operation to data-driven management, leading to optimized energy usage, lower maintenance costs, and enhanced operational control.

Market Challenges and Problem Identification

The Fruit and Vegetable Market of Santa Fe operates 87 warehouses, many of which rely on cold rooms to store perishable goods. However, these cold rooms face significant operational challenges due to outdated technology, inefficient energy usage, and lack of proper monitoring systems. These issues lead to high operational costs, environmental concerns, and reduced equipment lifespan. Below are detailed problems identified in the market.

Key Problems in the Market

High Energy Consumption

Why is this a problem?

Cold rooms operate continuously to maintain the required temperature for food preservation. However, without optimized cooling cycles and energy management, they consume excessive electricity, leading to high costs for operators.

Key Factors Contributing to High Energy Usage

- **Continuous Operation**: Cold rooms run 24/7 without dynamic adjustments, leading to unnecessary energy use.
- **No Load Management**: There is no system to adjust cooling intensity based on demand fluctuations.
- **Old Refrigeration Systems**: Many cold rooms use outdated refrigeration units that are not energy-efficient.
- **Lack of Insulation Monitoring**: Poor insulation leads to temperature loss, forcing cooling systems to work harder.
- **Peak Hour Consumption**: Electricity costs are higher during peak hours, but operators lack real-time data to shift energy usage to off-peak periods.

Impact

- Increased electricity bills for businesses.
- Higher operational costs, reducing profitability.
- More strain on the local power grid, leading to potential blackouts.

Reactive Maintenance Instead of Preventive Maintenance

Why is this a problem?

Currently, maintenance is **performed only when failures occur, which** leads to:

- **Unplanned Downtime:** Cold rooms may stop functioning suddenly, affecting stored goods.
- **Increased Repair Costs:** Emergency repairs are more expensive than scheduled maintenance.
- **Shortened Equipment Lifespan:** Without predictive monitoring, equipment wears out faster, requiring costly replacements.

Impact

- **Higher operational costs** due to frequent repairs.
- **Food spoilage risk**, leading to financial losses.
- **Inconsistent cooling performance**, affecting supply chain reliability.

Environmental Impact of Cold Rooms

Why is this a problem?

Cold rooms contribute significantly to carbon emissions, making them a concern from an environmental perspective. High energy consumption and poor refrigerant management lead to negative ecological effects.

Key Environmental Concerns

High Carbon Footprint: Excessive energy consumption leads to higher CO_2 emissions, increasing the market's environmental impact.
Refrigerant Leaks: Many cold rooms use fluorinated refrigerants (HFCs), which contribute to global warming if leaked into the atmosphere.

Wasted Energy Equals Wasted Resources: The more energy inefficient the system, the more fossil fuels are burned for electricity generation.

Food Waste Due to Temperature Issues: Improper cooling results in food spoilage, contributing to waste disposal problems.

Impact

- Increased greenhouse gas emissions, contributing to climate change.
- Higher operational costs due to inefficient energy use.
- Regulatory risks, as governments impose stricter energy and refrigerant laws.

Proposed Solution: Digitalization Using Asset Administration Shell (AAS)

To address the identified challenges, we propose the digitalization of cold rooms through the implementation of Asset Administration Shell (AAS). This approach will provide a structured, standardized digital representation of each cold room, enabling real-time monitoring, predictive maintenance, and energy optimization.

What is Asset Administration Shell (AAS)?

AAS is a digital framework that creates a virtual representation of a physical asset, such as a cold room. It allows for structured data collection, analysis, and integration with IoT-based monitoring systems.

By implementing AAS, we can store real-time operational data, historical maintenance records, and performance analytics in a centralized digital environment.

Key Benefits of Digitalization Using AAS

Energy Efficiency & Cost Reduction

- **Optimized Cooling Cycles:** Adjust cooling intensity based on real-time demand, reducing unnecessary power consumption.
- **Peak Load Management:** Shift energy-intensive operations to off-peak hours, lowering electricity costs.
- **Energy Usage Insights:** Track and analyze power consumption trends to identify inefficiencies.

Predictive Maintenance & Equipment Longevity

- **Real-Time Equipment Health Monitoring:** Detect early signs of failure in compressors, fans, and refrigeration units.
- **Maintenance Alerts:** AAS can trigger automatic maintenance reminders before failures occur.
- **Increased Equipment Lifespan:** Timely repairs prevent costly breakdowns and extend cold room durability.

Environmental Sustainability

- **Lower Carbon Footprint:** Reducing energy waste leads to lower CO_2 emissions.
- **Sustainable Cooling Strategies:** Optimize temperature settings to prevent energy overuse.
- **Compliance with Environmental Regulations:** Helps businesses meet energy efficiency standards and adopt green initiatives.

By adopting AAS-based digitalization, cold room operations can be transformed to become more efficient, cost-effective, and environmentally responsible. This structured approach ensures better energy management, predictive maintenance, and long-term sustainability in the Fruit and Vegetable Market of Santa Fe.

Requirements for Digitalization

To successfully digitalize cold rooms and make them more efficient, cost-effective, and manageable, we need specific hardware, software, and system requirements. Below is a detailed breakdown of what is needed:

Hardware Requirements (Physical Devices)

To collect, transmit, and store data, we need **smart sensors and computing devices**.

Sensors for Data Collection

These sensors are essential to measure different parameters inside the cold room:

- **Temperature Sensors** – Measure internal cooling temperature.
- **Humidity Sensors** – Monitor moisture levels.
- **Energy Meters** – Track power consumption of compressors and cooling units.
- **Door Status Sensors** – Detect whether the cold room door is open or closed.
- **Vibration Sensors** – Detect unusual vibrations in compressors, which can indicate failures.
- **Airflow Sensors** – Ensure proper circulation of cold air inside the room.

For temperature sensing, the use of the T-700 intelligent controller is proposed. This device not only enables temperature control within the cold room but also allows alarm configuration, features a real-time clock, and includes digital inputs and outputs.
Additionally, with its wireless internet connectivity, it supports real-time monitoring and event logging.
For energy measurement, the Cloud Energy Meter is proposed, an intelligent three-phase power analyzer with Wi-Fi connectivity.
This meter provides data on grid frequency, RMS voltage and current, power factor, and active, reactive, and apparent power, among other parameters.

Computing Devices (Edge Devices)

These devices **process and transmit sensor data** to a cloud or local database:

- **Microcontrollers** – Simple devices for collecting sensor data (e.g., Arduino, ESP32).

- **Single-Board Computers** – More powerful devices for local processing (e.g., Raspberry Pi, Jetson Nano).
- **Industrial IoT Gateways** – Securely transmit data to the cloud or a local storage system.

Communication Infrastructure

To ensure seamless data transfer, we need:

- **Wi-Fi/Ethernet** – For reliable network connectivity.
- **LoRaWAN** – For long-range, low-power communication if internet is not available.
- **MQTT Brokers** – To enable lightweight, real-time communication between sensors and the cloud.

Cloud & Storage

- **Cloud-based storage (AWS, Azure, Google Cloud)** – To store and analyze data remotely.
- **Local Database (SQL, InfluxDB)** – For on-site data storage if the internet is not always available.

Software Requirements

Software is needed to **process, store, visualize, and analyze** collected data.

Asset Administration Shell (AAS)

- The AAS acts as a digital representation of the cold room, storing all data in a structured format.
- It contains submodels such as:
  - **Operational Data** (temperature, humidity, energy use)
  - **Maintenance Data** (service history, expected failures)
  - **Technical Specifications** (power capacity, insulation details)

Data Collection & Processing Software

- **MQTT Protocol** – Enables real-time data transfer from sensors to the cloud.
- **OPC UA (Open Platform Communications Unified Architecture)** – Ensures interoperability between different IoT devices.

- **Edge Computing Frameworks** – For processing data at the source (e.g., Node-RED, Azure IoT Edge).

Dashboard & Visualization

- **Power BI / Grafana** – Used to create real-time monitoring dashboards.
- **Custom Web Application (React, Python Flask, or Node.js)** – If a user-friendly web dashboard is needed.
- **Mobile App (Optional)** – For remote monitoring of cold room conditions.

Functional Requirements

The system must be designed to **solve real problems** faced in cold room management.

Real-Time Monitoring

- The system should **collect live data** from temperature, humidity, and energy sensors.
- All **real-time data should be viewable on a dashboard** for operators.

Predictive Maintenance

- **Historical sensor data should be analyzed** to predict when a cooling unit may fail.
- Alerts should be **automatically triggered** when anomalies are detected.

Energy Optimization

- The system should **analyze energy consumption patterns** and suggest **ways to reduce electricity costs**.
- Load management features should **automatically adjust cooling cycles** based on need.

Security & Access Control

- **Data Encryption** – All data transmissions should be encrypted to prevent unauthorized access.
- **User Access Control** – Only authorized users should be able to modify system settings.
- **Automated Backups** – To prevent data loss in case of failures.

Summary of Requirements

Category	Requirement
Hardware	Temperature, humidity, energy, and airflow sensors; microcontrollers (ESP32, Raspberry Pi); communication devices (Wi-Fi, LoRaWAN).
Software	Asset Administration Shell (AAS); MQTT, OPC UA for communication; Power BI/Grafana for visualization.
Functional	Real-time monitoring, predictive maintenance, energy optimization, centralized data storage.
Security	Data encryption, user access control, automated backups.

Example Implementation: AAS for One Cold Room

For this AAS, a standard Cold Room used for storing fruits and vegetables was chosen as the model. However, depending on the specific configuration and type of Cold Room, the defined properties could be modified in the future to adapt to different operational, technical, or maintenance needs.

The AAS was designed using the AASX Package Explorer software, a tool that allows the creation and management of Asset Administration Shells (AAS) according to Industry 4.0 standards.

Within the AAS, submodels have been defined, which are organized structures that group information related to different aspects of the Cold Room, such as identification, technical data, operation, and maintenance.

Each submodel contains properties that represent specific system data, whether fixed values, configurable parameters, or real-time variables. These elements enable a digitalized and efficient asset management process.

The following sections describe each submodel.

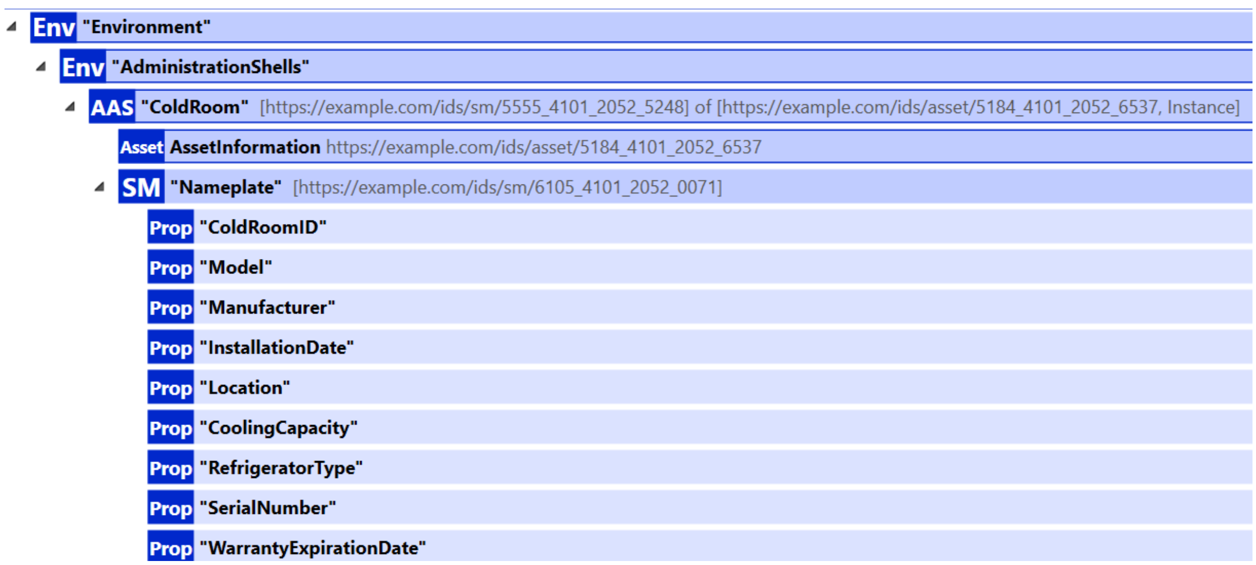
Nameplate Submodel

This submodel contains the basic and static information of the Cold Room. Its purpose is to facilitate equipment identification and provide essential data for its management.

Properties:

- ColdRoomID: Unique identifier of the Cold Room.
- Model: Equipment model.

- Manufacturer: Manufacturer’s name.
- InstallationDate: Date the equipment was installed.
- Location: Physical location within the market.
- CoolingCapacity: Cooling capacity in kW or BTU/h.
- RefrigeratorType: Type of refrigeration system (vapor compression, absorption, etc.).
- SerialNumber: Equipment’s serial number, useful for traceability and technical support.
- WarrantyExpirationDate: Manufacturer’s warranty expiration date.



TechnicalData Submodel

This submodel groups the technical characteristics of the refrigeration system. Its data allows for performance evaluation and energy or maintenance analysis.

Properties:

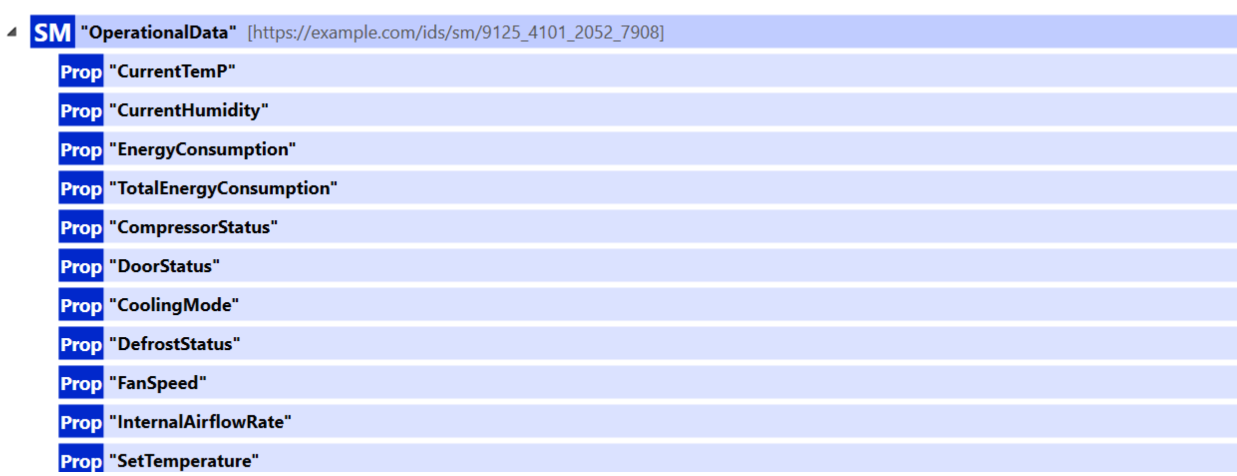
- CoolingSystem: Type of refrigeration system (direct, indirect, etc.).
- EvaporatorType: Type of evaporator used (plate, tube, finned).
- CompressorType: Type of compressor (piston, scroll, screw).
- InsulationTechnic: Insulation technique used in the Cold Room.
- ThermalConductivityOfInsulation: Thermal conductivity of the insulation (W/m·K).
- MaxEnergyDemand: Maximum energy demand in kW.
- RefrigerantFlowRate: Refrigerant flow rate in kg/h.
- OperatingVoltage: System operating voltage.
- HeatExchangerEfficiency: Heat exchanger efficiency.
- OperatingPressure: Refrigeration system operating pressure.

OperationalData Submodel

This submodel stores real-time operational data of the Cold Room. These values allow monitoring of the equipment's status and making adjustments if necessary.

Properties:

- CurrentTemp: Current temperature inside the Cold Room (°C).
- CurrentHumidity: Relative humidity level inside (%).
- EnergyConsumption: Instantaneous energy consumption in kW.
- TotalEnergyConsumption: Total accumulated energy consumption in kWh.
- CompressorStatus: Compressor status (on/off).
- DoorStatus: Door status (open/closed).
- CoolingMode: Active cooling mode (eco, fast, etc.).
- DefrostStatus: Defrost cycle status (active/inactive).
- FanSpeed: Fan speed in RPM.
- InternalAirflowRate: Airflow rate inside the Cold Room in m^3/h.
- SetTemperature: Configured target temperature.

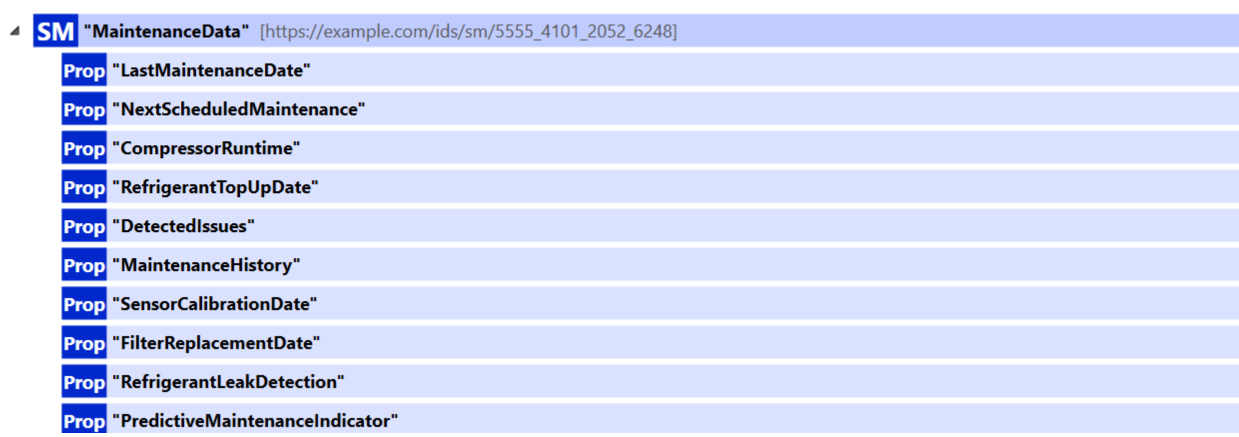


MaintenanceData Submodel

This submodel records information about system maintenance, including intervention history and scheduling of future services.

Properties:

- LastMaintenanceDate: Date of the last performed maintenance.
- NextScheduledMaintenance: Date of the next scheduled maintenance.
- CompressorRuntime: Accumulated operating hours of the compressor.
- RefrigerantTopUpDate: Last date the refrigerant was refilled.
- DetectedIssues: Detected issues in the system.
- MaintenanceHistory: Record of performed maintenance activities.
- SensorCalibrationDate: Date of the last sensor calibration.
- FilterReplacementDate: Date the system filter was replaced.
- RefrigerantLeakDetection: Indication of refrigerant leak presence.
- PredictiveMaintenanceIndicator: Data-based indicator suggesting the need for preventive maintenance.



Future Work and Implementation Roadmap

The future work and implementation roadmap outlines the steps required to fully integrate digitalization into the cold rooms at the Fruit and Vegetable Market of Santa Fe. Since this project proposal primarily focuses on problem identification and solution planning, the actual execution of the digitalization strategy will be part of future phases.

Step 1: Pilot Implementation (Testing on One Cold Room)

Before scaling the solution to all cold rooms, we need to **test the system** on **one selected cold room**. This will help in identifying **practical challenges, fine-tuning the system, and ensuring feasibility**.

Tasks in this Step:

- Select a **cold room** to be used for the pilot project.
- Gather **basic details** such as size, energy usage, and cooling system specifications.
- Install **basic sensors** (temperature, humidity, and energy consumption meters).
- Connect the **sensors to a data collection system** (using IoT devices like Raspberry Pi or industrial controllers).
- Store **data in a simple database** for analysis.

Step 2: IoT Integration and Data Collection

After the pilot project is set up, we will add more IoT components to enable full monitoring.

Tasks in this Step:

- Install additional sensors (such as door open/close status and airflow sensors).
- Configure a real-time data transmission system using MQTT or OPC UA protocols.
- Store all collected data in a centralized cloud or local storage system.
- Set up alerts and notifications for abnormal temperature or energy consumption.

Step 3: Cloud-Based AAS Deployment

Once data collection is stable, we will create a digital twin of the cold room using Asset Administration Shell (AAS). This will store structured information about the cold room, including real-time data, maintenance history, and operational conditions.

Tasks in this Step:

- Develop the AAS model with all necessary submodels (Nameplate Data, Operational Data, Maintenance Data, Technical Data).
- Deploy the AAS on a cloud platform like AWS, Azure, or a local server.
- Set up API access so the AAS can be integrated with other systems (such as dashboards or mobile apps).

- Validate data flow between sensors, AAS, and cloud storage.

Step 4: Dashboard Development for Visualization

To make the collected data easy to understand, a dashboard will be developed using Power BI or Grafana.

Tasks in this Step:

- Design a user-friendly interface that shows real-time and historical trends.
- Include key metrics such as temperature, energy usage, maintenance logs.
- Implement alerts and reports for unusual conditions (e.g., sudden temperature rise).
- Test dashboard performance and improve based on user feedback.

Step 5: Scaling Across the Market

Once the pilot project is successful, the same digitalization approach will be expanded to all cold rooms in the market.

Tasks in this Step:

- Identify additional cold rooms that need digitalization.
- Adjust the AAS structure to accommodate different cold room types (old vs. new models).
- Standardize hardware and software configurations to ensure consistency.
- Provide training to market staff on how to use the system.

Step 6: Renewable Energy Integration

To further reduce costs and improve sustainability, cold rooms can be powered using renewable energy sources.

Tasks in this Step:

- Explore solar panel installation to supply power to cold rooms.
- Investigate the use of biodigesters to convert waste into energy.
- Connect renewable energy systems to the existing AAS-based monitoring system.

Final Expected Outcomes

By following this implementation roadmap, we expect the following benefits:

- **Lower Energy Costs**: Reduction of at least 15-20% in energy bills through optimized cooling cycles.
- **Operational Efficiency**: Real-time monitoring enables better resource allocation and maintenance scheduling.
- **Scalability**: The system can be expanded to any number of cold rooms.
- **Sustainability**: Using renewable energy solutions will further reduce carbon footprint.

Conclusion

Cold rooms play a vital role in preserving perishable goods in the Fruit and Vegetable Market of Santa Fe. However, the market faces significant challenges due to the lack of real-time monitoring, excessive energy consumption, and inefficient maintenance practices. These issues lead to higher operational costs, equipment failures, and unnecessary energy wastage.

Through our study, we identified that digitalizing cold rooms can help overcome these challenges. The introduction of Asset Administration Shell (AAS) provides a structured way to collect, organize, and analyze critical operational data, making it easier for businesses to optimize energy usage and improve overall efficiency.

With AAS, cold room operators can monitor key parameters such as temperature, humidity, power consumption, and equipment status in real-time. This allows for better decision-making, enabling operators to identify inefficiencies, reduce downtime, and optimize cooling cycles to lower energy costs.

Another major benefit of digitalization is predictive maintenance. Instead of waiting for equipment to break down, sensor-based monitoring and historical data analysis can help predict failures before they happen. This reduces repair costs, prevents sudden breakdowns, and ensures smooth operations without disruptions.

While this report presents a proposal for how digitalization can solve existing problems, the actual implementation will be a future step. As a starting point, we have demonstrated how AAS can be applied to one cold room as an example. The next phase of this project would involve deploying IoT sensors, integrating cloud-based monitoring, and expanding AAS implementation to multiple cold rooms.

Key Takeaways

- Digitalization through AAS will help monitor and manage cold rooms more efficiently.
- Energy consumption can be optimized, reducing unnecessary costs.
- Predictive maintenance can help prevent unexpected breakdowns.
- Standardized data collection will allow for long-term improvements and benchmarking.
- This proposal serves as the foundation for future implementation and scalability.

By implementing these digital solutions, the market can transition to a more sustainable, cost-effective, and efficient way of managing cold storage operations.